

Lecture 1

What is a PDE?

- Roughly: an equation involving partial differentiation of a function

- Examples: $\partial_t^2 u(t, x) - \partial_x^2 u(t, x) = 0$

$$\partial_t u(t, x) - u(t, x) \partial_x u(t, x) = 0$$

$$\partial_x^2 u(x, y) + \partial_y^2 u(x, y) = 0$$

Remarks: - The unknown is a function

- An ODE takes the form

$$\frac{d}{dt} u(t) = F(u)$$

u is a function of 1 variable, a PDE involves functions of several variables

- Typically comes w/ additional info called "data," e.g. boundary value, initial value $\subset \mathbb{R}^d$

More precisely: a PDE for a function $u: \Omega \rightarrow \mathbb{R}$
e.g. $u(x)$, $x = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \in \Omega$

$$F\left(x, u(x), \frac{\partial u}{\partial x_i}(x), \dots, \frac{\partial^m u}{\partial x_{i_1} \dots \partial x_{i_m}}\right) = 0, \quad x \in \Omega \quad (\otimes)$$

- m is called the order of the PDE

- a classical solution admits continuous partial derivatives up to order m and satisfies $(\otimes)$ for every $x \in \Omega$

- u can also be vector valued, e.g. $\vec{u}(x) = \begin{pmatrix} u_1(x) \\ \vdots \\ u_n(x) \end{pmatrix}$

Remarks: - Classical solution often too strong a requirement, "weak solutions" will be useful

- The equation is local, every object in the equation is evaluated at the same point

Why study PDEs?

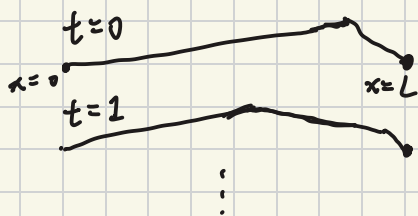
- Ubiquitous in Science, Engineering, and Math.

- ex. - Electricity and Magnetism (Maxwell's eq)
- Quantum Mechanics (Schrödinger's eq)
- Fluids (Navier-Stokes eq)
- Cosmology (Einstein's eq)

Many natural phenomena are local!!

- behavior at a point depends only on "nearest neighbor" and its location

Ex. Wave equation

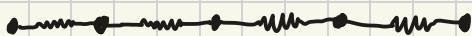


described by

$$\partial_t^2 u - \partial_x^2 u = 0$$

$$u(t, 0) = u(t, L) = 0$$

Can approximate by masses and springs



Forces on a mass determined only by its displacement and the displacement of its nearest neighbors.

Types of equations

There is no general theory of PDEs that allows us to analyze any PDE, so we restrict our attention to several useful classes

Def. a PDE is called linear if

$$P u = f \quad (\text{**)}$$

where $f: \Omega \rightarrow \mathbb{R}$ is independent of u and P is a differential operator. We will only consider order 1 and 2 equations, so

$$P = - \sum_{j,k=1}^n a_{j,k} \frac{\partial^2}{\partial x_j \partial x_k} + \sum_{j=1}^n b_j \frac{\partial}{\partial x_j} + c$$

where $a_{j,k}, b_j, c: \Omega \rightarrow \mathbb{R}$ are given coefficients

Since $\frac{\partial^2}{\partial x_j \partial x_k} = \frac{\partial^2}{\partial x_k \partial x_j}$, we can always arrange s.t.

$a_{j,k} = a_{k,j}$, i.e. $[a_{j,k}]$ is a symmetric matrix

(**) is linear in the sense that P is a linear operator: for $u, v \in C^\infty(\Omega)$, $\lambda \in \mathbb{R}$

$$P u + P v = P(u+v)$$

$$P(\lambda u) = \lambda P u$$

Def. (**) is called elliptic if the matrix of coefficients $[a_{j,k}]$ is positive definite, meaning all eigenvalues of $[a_{j,k}]$ is strictly positive.

Prototype of an elliptic operator:

$$L = -\Delta, \quad \Delta := \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2} = \frac{\partial^2}{\partial x_1^2} + \dots + \frac{\partial^2}{\partial x_n^2}$$

$\leadsto$ Laplace's equation: $\Delta u = f$

Physical interpretation: u = electric potential
 f = charge density.

- Often there will be a distinguished time variable that behaves differently than spatial variables, these are called evolution equations
- Two classic types of evolution equations called hyperbolic and parabolic

hyperbolic: $P = \partial_t^2 - \sum \alpha_{jk} \partial_{x_j} \partial_{x_k} + \text{l.o.t.}$

parabolic: $P = \partial_t - \sum \alpha_{jk} \partial_{x_j} \partial_{x_k} + \text{l.o.t.}$

$[\alpha_{jk}]$ positive definite

We will consider two prototypes:

1.) $P = \partial_t^2 - \Delta$ (wave operator)

2.) $P = \partial_t - \Delta$ (heat operator)

Examples of ill-posedness

1.) Global well-posedness for ODEs (failed existence)

$$\frac{d}{dt} u(t) = 1 + u^2, \quad u(0) = 0$$

$$\text{separable} \leadsto \int_0^u \frac{1}{1+u^2} du = \int_0^t dt$$

$$\arctan u(t) = t$$

$$u(t) = \tan(t)$$

$\leadsto$, blows up as $t \rightarrow \frac{\pi}{2}$

2.) No Dirichlet data (violates uniqueness)

$\Delta u = 0$ in Ω has many solutions

2.) Backward heat equation (violates continuous dep.)

$$(t, x) \in (-\infty, 0] \times \mathbb{R}^n$$

$$\begin{cases} (\partial_t - \Delta) u = 0 \\ u(0, x) = f(x) \end{cases}$$

Intuition: heat wants to diffuse, smooths out roughness, turns out this roughness decays exponentially in time, so backwards in time, roughness blows up exponentially, which violates continuous dependence.